# Final Exam Review

Module 8 - Nonlinear Pharmacokinetics · Review set · 19 Oct 2026

Name _____ Date _____

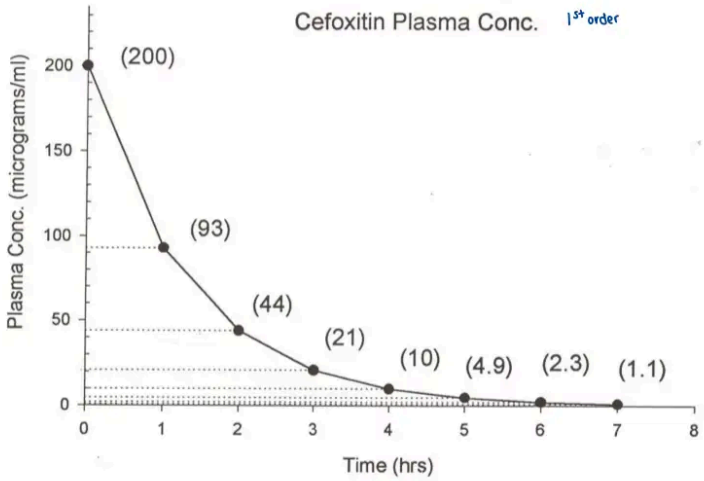
**Slide 3** Which of the following is characteristic of first-order processes?

A. units of k of (concentration or amount)/time
B. a constant rate of elimination
C. dC/dt = -k
D. a constant half-life

**Slide 7** 400 mg of a drug was administered by IV bolus injection. The equation describing the drug in the body is given as $C_p = (40\ mg/L)e^{-0.9t}$. What is the apparent volume of distribution of this drug?

A. 10 L
B. 40 L
C. 20 L
D. 50 L

**Slide 8** Cefoxitin plasma concentrations were 200, 93, 44, 21, 10, 4.9, 2.3 and 1.1 micrograms/mL at 0, 1, 2, 3, 4, 5, 6 and 7 hours, as shown. Estimate the half-life of cefoxitin.




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**Slide 10** The renal clearance of an antibiotic was estimated to be 45 mL/min. What can be inferred from this information?

A. The drug is filtered and reabsorbed.

B. The drug is filtered and actively secreted.

C. There is some evidence of renal impairment.

**Slide 11** Hepatic clearance is most readily approximated with creatinine clearance.

A. False

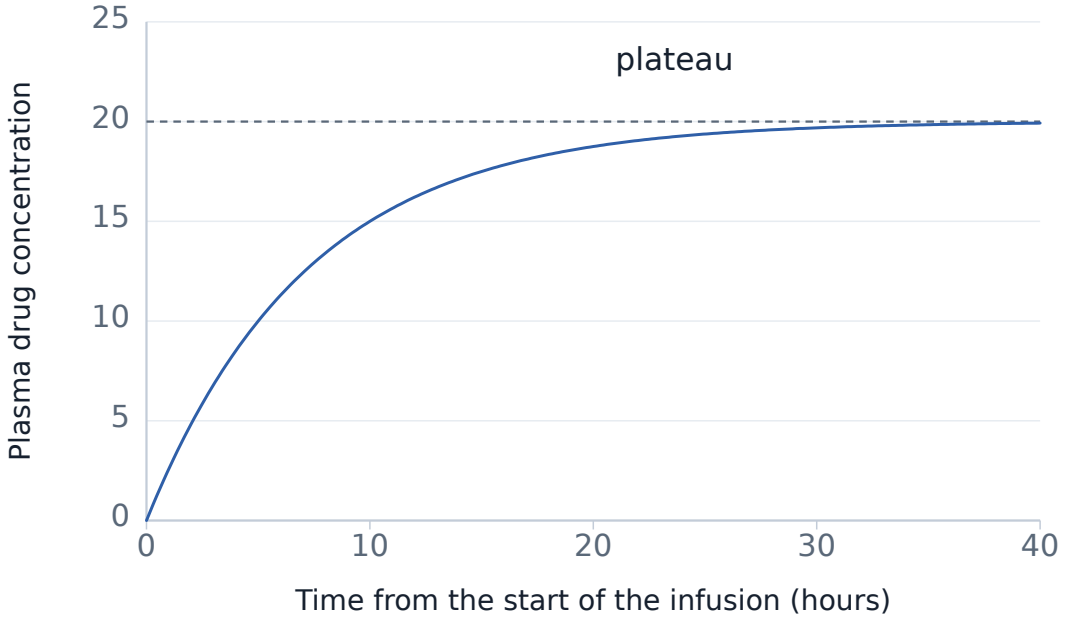
B. True

**Slide 13** What is the expected maximum steady state plasma concentration following multiple IV bolus dosing every 12 hours of 300 mg of a drug with a t½ of 4 hours and $V_D$ of 16 L?

A. None of the above
B. 32.6 mg/L
C. 21.4 mg/L
D. 18.7 mg/L

**Slide 13** A drug with a t½ of 4 hours and $V_D$ of 16 L is given as multiple IV bolus doses of 300 mg every 12 hours. What is the minimum plasma concentration at steady state?

**Slide 15** Which type of drug input is depicted by this curve?




A. instantaneous input
B. zero-order input
C. first-order input

**Slide 15** Match each pattern of drug input and output to the type of administration that gives it.

1. Instantaneous input, first-order output _____
2. First-order input, first-order output _____
3. Zero-order input, first-order output _____

A. IV bolus
B. Conventional oral dose
C. IV infusion

**Slide 17** What is the bioavailability of a 500 mg tablet which yielded an AUC of 115 (mg/L)hr compared to a 400 mg IV bolus dose that yielded an AUC of 132 (mg/L)hr?

A. 50%
B. 92%
C. 70%
D. 85%

**Slide 17** A 500 mg tablet yielded an AUC of 115 (mg/L)hr and a 400 mg IV bolus dose yielded an AUC of 132 (mg/L)hr. What oral dose would provide similar therapeutic activity as the 400 mg bolus dose? Round to the nearest 5.

**Slide 19** Drugs that display Michaelis-Menten saturation kinetics will often exhibit zero-order kinetics at low concentrations and first-order kinetics at high concentrations.

A. False
B. True

**Slide 21** The trapezoidal rule provides a simple way to estimate the AUC.

A. False
B. True

**Slide 23** What is the duration of action for a 1 g IV bolus dose with a $V_D$ = 12.5 L, t½ = 4hr and MEC= 20 mg/L ?

A. None of the above
B. 4 hours
C. 12 hours
D. 8 hours

**Slide 23** A drug has a $V_D$ = 12.5 L, t½ = 4hr and MEC= 20 mg/L. If 500 mg were given as an IV bolus dose instead of 1 g, what would the duration of action be?

**Slide 25** What is the plasma concentration at the end of a 2 hour infusion of a drug with a t½ of 3 hours and a $V_D$ of 18 L at a rate of 50 mg/hr?

A. 3.03 mg/L
B. 4.45 mg/L
C. 4.81 mg/L
D. 31.5 mg/L

**Slide 25** A drug with a t½ of 3 hours and a $V_D$ of 18 L is infused at a rate of 50 mg/hr for 2 hours. What is the concentration 3 hours after cessation of the infusion?

**Slides 25 and 26**

A drug with a t½ of 3 hours and a $V_D$ of 18 L is infused at a rate of 50 mg/hr for 2 hours. Another 2 hour infusion was started 6 hours after the start of the first infusion. What is the concentration 4 hours after cessation of the second infusion?

**Slide 25** A drug with a t½ of 3 hours and a $V_D$ of 18 L is infused at a rate of 50 mg/hr. If the infusion were continued at this rate, what would the steady-state concentration be?

**Slide 25** A drug with a t½ of 3 hours and a $V_D$ of 18 L is infused. If it were infused at a rate of 100 mg/hr, what would the steady-state concentration be?

**Slide 25** A drug with a t½ of 3 hours infused at 50 mg/hr reaches a steady-state concentration of about 12 mg/L. Infused at 100 mg/hr, it reaches about 24 mg/L. How long does it take to reach 24 mg/L at 100 mg/hr, compared with reaching 12 mg/L at 50 mg/hr?

A. Twice the time
B. Half the time
C. The same time

# Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

## Slide 3 Worked by the drill

**Answer:** D. a constant half-life

For a first-order process $t½ = \frac{0.693}{k}$, and k is a constant, so the half-life does not depend on how much drug is present. The time to go from 100 to 50 is the same as from 10 to 5.

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## Slide 7 Worked by the drill

**Answer:** A. 10 L

### THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = C_0e^{-kt}$, so $C_0$ = 40 mg/L and k = 0.9 $hr^{-1}$

The one-compartment IV bolus equation. At t = 0 the exponential is 1, so the coefficient is the concentration just after the dose.

ALGEBRA

$$V_D = \frac{D_0}{C_0} = \frac{400\ mg}{40\ mg/L} = 10\ L$$

The apparent volume relates the amount in the body to the plasma concentration, and just after a bolus the amount in the body is the whole 400 mg dose. mg divided by mg/L leaves L.

For an IV bolus $C_p = C_0e^{-kt}$, so the number in front of the exponential is $C_0$, 40 mg/L. $V_D = \frac{D_0}{C_0} = \frac{400\ mg}{40\ mg/L} = 10$ L.

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## Slide 8 Worked by the drill

**Answer:** 1 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **C at 1 hr** | 93 | the level near 100 on the curve |
| **C at 2 hr** | 44 | about half of 93, one hour later |
| **$C_0$** | 200 | the first point; 93 one hour later is about half of it |

### THE WORKING, ONE LINE AT A TIME

SETUP

t½ is the time for the concentration to fall to half: from 93 at 1 hr to 44 at 2 hr, about half

Half-life is read straight off the curve as the time between a level and half of it. Half of 100 is roughly 50, and the points near those levels are 93 and 44, one hour apart.

ALGEBRA

$\frac{93}{44} = 2.11$ and $\frac{44}{21} = 2.10$, so each hour the level falls to about half

Every one-hour step divides the concentration by about 2, whatever the starting level. A constant fractional fall is first order, so the half-life is the same all along the curve.

ALGEBRA

$k = \ln(\frac{93}{44}) = 0.748\ hr^{-1}$ over the 1-hour step; $t½ = \frac{0.693}{0.748} = 0.93$ hr

The same reading from two points: ln of the ratio over the 1 hour between them is k, and $t½ = \frac{0.693}{k}$. The curve on a linear axis is curved; on a log axis these points fall on a straight line.

ROUND

$t½ \approx 1$ hr

Halving by eye gives about 1 hour, her estimate; the points give 0.93 hr. Both read the same half-life to one significant figure.

> **DOES THE NUMBER MAKE SENSE?**
> Each one-hour step roughly halves the level, 200 to 93 to 44 to 21, so the half-life is close to 1 hr; the points give 0.93 hr.

Halving by eye gives about 1 hour; k from the points gives 0.93 hr. Either is accepted.

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## Slide 10 Worked by the drill

**Answer:** A. The drug is filtered and reabsorbed.

### THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_R = 45$ mL/min; GFR $\approx 120$ mL/min

Renal clearance is compared with the glomerular filtration rate, the plasma that filtration alone clears each minute.

ALGEBRA

$\frac{45\ mL/min}{120\ mL/min} = 0.375$, so $Cl_R$ is less than GFR: filtration with tubular reabsorption

Below the filtration rate means drug that was filtered is not all leaving in the urine: some is reabsorbed back into the blood.

45 mL/min is below the glomerular filtration rate of about 120 mL/min. The drug is filtered, and some of it returns from the tubule to the blood, so less plasma is cleared than is filtered.

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## Slide 11 Worked by the drill

**Answer:** A. False

Hepatic clearance is approximated by difference: $Cl_H = Cl_T - Cl_R$, total clearance minus renal clearance. Creatinine clearance, from Cockcroft-Gault, estimates the patient's renal function instead.

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## Slide 13 Worked by the drill

**Answer:** C. 21.4 mg/L

### THE WORKING, ONE LINE AT A TIME

UNIT

$$k = \frac{0.693}{4\ \text{hr}} = 0.17325\ \text{hr}^{-1};\ C_0 = \frac{300\ \text{mg}}{16\ \text{L}} = 18.75\ \text{mg/L}$$

The rate constant from the half-life, and the first-dose peak from dose over volume: for an IV bolus the whole dose is in the body at once.

ALGEBRA

$$k\tau = 0.17325 \times 12 = 2.079;\ e^{-2.079} = 0.1251$$ (12.5% of a dose left after 3 half-lives)

$\tau$ is the dosing interval, 12 hr. $k\tau$ has no units, and $e^{-k\tau}$ is the fraction of each dose still in the body when the next is given.

ALGEBRA

$$C_{max}^{\infty} = \frac{C_0}{1 - e^{-k\tau}} = \frac{18.75}{1 - 0.1251} = \frac{18.75}{0.8749} = 21.43\ \text{mg/L}$$

The principle of superposition: the first-dose peak is carried to steady state by the accumulation factor, $\frac{1}{0.8749} = 1.14$, because each dose adds to what remains.

ROUND

$$C_{max}^{\infty} = 21.4\ \text{mg/L}$$

21.43 mg/L to one decimal place, her value. It is the steady-state peak, reached just after each dose is given, and it is 1.14 times the first-dose peak of 18.75 mg/L.

$C_0 = 18.75$ mg/L, $k = 0.17325\ \text{hr}^{-1}$, and $e^{-k\tau} = 0.1251$ for $\tau = 12$ hr. $C_{max}^{\infty} = \frac{18.75}{1 - 0.1251} = 21.43$ mg/L.

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## Slide 13 Worked by the drill

**Answer:** 2.7 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$C_{max}^{\infty}$** | 21.4 mg/L | part a: the peak the trough falls from |
| **t½** | 4 hours | $k = 0.17325\ \text{hr}^{-1}$ |
| **$\tau$** | every 12 hours | the time from peak to trough |
| **$V_D$** | 16 L | *not needed once the peak is known* |
| **$D_0$** | 300 mg | *not needed once the peak is known* |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min}^{\infty} = C_{max}^{\infty}\, e^{-k\tau}$$

The trough at steady state is the peak after one dosing interval of first-order decline: it comes at the end of the interval, just before the next dose.

UNIT

$$k = \frac{0.693}{4\ \text{hr}} = 0.17325\ \text{hr}^{-1};\ k\tau = 0.17325 \times 12 = 2.079;\ e^{-2.079} = 0.1251$$

$\tau$ = 12 hr, the dosing interval, which is 3 half-lives of 4 hours. $\text{hr}^{-1} \times \text{hr}$ cancels, so the exponent is a pure number.

ALGEBRA

$$C_{min}^{\infty} = 21.4\ \text{mg/L} \times 0.1251 = 2.677\ \text{mg/L}$$

The peak from part a times the fraction left after 12 hours. Carrying the peak unrounded, 21.43 × 0.1251 = 2.681 mg/L.

ROUND

$$C_{min}^{\infty} = 2.7\ \text{mg/L}$$

2.677 mg/L to one decimal place, her value. The 12-hour interval is three half-lives of 4 hours, and three half-lives leave one eighth of the 21.4 mg/L peak.

> **DOES THE NUMBER MAKE SENSE?**
> The trough is the peak times the fraction left after 12 hours, 3 half-lives: one eighth of 21.4 mg/L is about 2.7 mg/L.

## Slide 15 Worked by the drill

**Answer:** B. zero-order input

Zero order means a constant rate of input: an IV infusion. With first-order output, the level rises until the rate out equals the rate in and then holds at the steady-state level.

## Slide 15 Worked by the drill

**Answer:** Instantaneous input, first-order output → IV bolus; First-order input, first-order output → Conventional oral dose; Zero-order input, first-order output → IV infusion

### EACH PAIR, AND WHY

| | |
|---|---|
| **Instantaneous input, first-order output** | IV bolus<br>The whole dose enters the blood at once, so there is no input phase on the curve: it starts at $C_0$ and falls by first-order elimination. |
| **First-order input, first-order output** | Conventional oral dose<br>Absorption from the gut is first order: fast while much drug is at the absorption site, slower as it empties. The curve rises to a peak and then falls by first-order elimination. |
| **Zero-order input, first-order output** | IV infusion<br>The pump delivers a constant rate, which is zero order. Against first-order elimination the level rises until rate out equals rate in, the steady state. |

## Slide 17 Worked by the drill

**Answer:** C. 70%

### THE WORKING, ONE LINE AT A TIME

SETUP

$$F_{abs} = \left(\frac{AUC_{po}}{AUC_{IV}}\right)\left(\frac{D_{IV}}{D_{po}}\right)$$

Absolute bioavailability compares the tablet with the IV bolus, so the IV AUC is the denominator. The doses differ, so the dose ratio corrects for them with the IV dose on top.

ALGEBRA

$$\frac{115}{132} = 0.871;\ \frac{400}{500} = 0.8;\ F = 0.871 \times 0.8 = 0.697$$

Absolute bioavailability is the AUC ratio scaled by the dose ratio. The units cancel in both ratios, (mg/L)hr over (mg/L)hr and mg over mg, so F is dimensionless.

ROUND

F = 0.697, which is 70%

0.697 rounds to 0.70, written as 70% or 0.70, never .70. F is a fraction with no units; below 1 it means about 30% of the oral dose does not reach the systemic circulation.

$F = \left(\frac{AUC_{po}}{AUC_{IV}}\right)\left(\frac{D_{IV}}{D_{po}}\right) = \left(\frac{115}{132}\right)\left(\frac{400}{500}\right) = 0.871 \times 0.8 = 0.697$, which is 70%. The IV AUC is in the denominator; the IV dose is on top of the dose ratio.

## Slide 17 Worked by the drill

**Answer:** 570 mg

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_{IV}$** | 400 mg bolus dose | the numerator of $D_{po} = D_{IV} \div F$ |
| **F** | 70% | part a: the denominator, 0.70 |
| **$D_{po}$** | 500 mg tablet | *not needed: the tablet strength gave F, not the new dose* |
| **AUC** | 115 (mg/L)hr | *not needed once F is known* |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$D_{po} = \frac{D_{IV}}{F}$$

Similar activity means the same AUC, the same amount reaching the circulation. Setting the two AUCs equal in the bioavailability equation leaves $F = \frac{D_{IV}}{D_{po}}$, rearranged for the oral dose.

ALGEBRA

$$D_{po} = \frac{400\ mg}{0.70} = 571.43\ mg$$

F = 70% from part a, written 0.70. Only 70% of an oral dose reaches the circulation, so the oral dose must be larger than the 400 mg IV dose.

ROUND

$D_{po}$ = 570 mg, to the nearest 5

571.43 is nearer 570 than 575. With F unrounded, $\frac{400}{0.697}$ = 573.89 mg, which is 575 to the nearest 5; the tolerance accepts both.

> **DOES THE NUMBER MAKE SENSE?**
> F is below 1, so the oral dose must exceed the 400 mg IV dose: 400 ÷ 0.70 gives 571.43 mg, which is 570 to the nearest 5.

With F unrounded, 0.697, the dose is 573.9 mg, which is 575 to the nearest 5. Her answer uses F = 70% and gives 570 mg. Both are accepted.

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## Slide 19 Worked by the drill

**Answer:** A. False

The orders are the wrong way round. At low concentrations elimination looks first order; at high concentrations the process proceeds at Vmax, a constant for a given drug in a given patient, which is zero order.

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## Slide 21 Worked by the drill

**Answer:** B. True

The trapezoidal rule breaks the curve into segments between the measured points, finds the area of each, one half base times height summed over the two sides, and adds them. It estimates the AUC from the data alone.

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## Slide 23 Worked by the drill

**Answer:** D. 8 hours

### THE WORKING, ONE LINE AT A TIME

UNIT

$$1\text{ g} = 1000\text{ mg};\ C_0 = \frac{1000\text{ mg}}{12.5\text{ L}} = 80\text{ mg/L};\ k = \frac{0.693}{4\text{ hr}} = 0.17325\text{ hr}^{-1}$$

The concentration just after the bolus is the dose over the volume, in the same units as the MEC. k comes from the half-life.

ALGEBRA

$$t_{eff} = \frac{\ln C_0 - \ln MEC}{k} = \frac{\ln(80 \div 20)}{0.17325} = \frac{1.3863}{0.17325} = 8.00\text{ hr}$$

$C = C_0 e^{-kt}$ solved for the time the level reaches the MEC. 80 ÷ 20 = 4, two doublings, so the duration is two half-lives.

ROUND

$t_{eff}$ = 8 hours

8.00 hr is two half-lives of 4 hours, 80 to 40 to 20 mg/L. The effect lasts while the level is at or above the MEC of 20 mg/L, so the duration of action is 8 hours.

$C_0 = \frac{1000 \text{ mg}}{12.5 \text{ L}}$ = 80 mg/L, four times the MEC. The level halves every 4 hours, so 80 to 40 to 20 mg/L takes two half-lives: 8 hours.

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## Slide 23 Worked by the drill

**Answer:** 4 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 500 mg | $C_0 = 500 \div 12.5 = 40$ mg/L |
| **$V_D$** | 12.5 L | divides the dose to give $C_0$ |
| **t½** | 4hr | $k = 0.17325$ hr$^{-1}$ |
| **MEC** | 20 mg/L | the level where the effect ends |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{500 \text{ mg}}{12.5 \text{ L}} = 40 \text{ mg/L}; \; k = \frac{0.693}{4 \text{ hr}} = 0.17325 \text{ hr}^{-1}$$

For an IV bolus $C_0$ is the dose over the volume, and mg over L leaves mg/L, the units of the MEC. Half the dose gives half the starting concentration, because the volume does not change with the dose.

ALGEBRA

$$t_{eff} = \frac{\ln(40 \div 20)}{0.17325} = \frac{0.6931}{0.17325} = 4.00 \text{ hr}$$

The natural-log duration line with the new $C_0$. 40 mg/L is twice the MEC, so the level reaches the MEC after one half-life.

ROUND

$t_{eff} = 4$ hours

Halving the dose takes one half-life, 4 hours, off the 8 hours of part a. The dose sits inside the logarithm, so half the dose does not give half the duration in general; here it does only because 8 hours is two half-lives.

> **DOES THE NUMBER MAKE SENSE?**
> $C_0$ of 40 mg/L is twice the MEC of 20 mg/L, so the level falls to the MEC in one half-life, 4 hr.

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## Slide 25 Worked by the drill

**Answer:** B. 4.45 mg/L

### THE WORKING, ONE LINE AT A TIME

UNIT

$$k = \frac{0.693}{3 \text{ hr}} = 0.231 \text{ hr}^{-1}; \; Cl = kV_D = 0.231 \times 18 = 4.158 \text{ L/hr}$$

No clearance is given, so it is built from the half-life and the volume: k is 0.693 over t½ for a first-order process, and $Cl = kV_D$. hr$^{-1}$ × L leaves L/hr.

ALGEBRA

$$\frac{R}{Cl} = \frac{50 \text{ mg/hr}}{4.158 \text{ L/hr}} = 12.025 \text{ mg/L}$$

At steady state the rate in, R, equals the rate out, Cl × $C_{ss}$, so $C_{ss}$ is R over Cl, the level the infusion climbs toward. mg/hr over L/hr leaves mg/L; the hours cancel.

ALGEBRA

$$1 - e^{-0.231 \times 2} = 1 - e^{-0.462} = 1 - 0.6300 = 0.3700$$

During an infusion $C_p = C_{ss}(1 - e^{-kt})$, so $1 - e^{-kt}$ is the fraction of steady state reached after 2 hours. $hr^{-1}$ × hr cancels, so kt and the fraction have no units.

ALGEBRA

$$C_p = 12.025 \times 0.3700 = 4.449 \text{ mg/L}$$

$C_p = C_{ss}(1 - e^{-kt})$: the steady-state level times the fraction reached. mg/L times a pure number stays mg/L. The answer must lie below 12.025 mg/L, and it does.

ROUND

$$C_p = 4.45 \text{ mg/L}$$

4.449 mg/L to two decimal places, her value. Two hours is less than one half-life of 3 hours, so the level is still below half of the 12.025 mg/L steady state.

$C_{ss} = \frac{R}{kV_D} = \frac{50}{0.231 \times 18} = 12.025$ mg/L, and after 2 hours the level is $12.025 \times (1 - e^{-0.462}) = 12.025 \times 0.3700 = 4.449$ mg/L.

## Slide 25 Worked by the drill

**Answer:** 2.2 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **Cend** | 4.45 mg/L | part a: the level the decline starts from |
| **t½** | 3 hours | k = 0.231 $hr^{-1}$ |
| **t** | 3 hours after cessation | one half-life |
| **R** | 50 mg/hr | *not needed once the end level is known* |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = \text{Cend } e^{-kt}\text{, t counted from the end of the infusion}$$

Once the infusion ends no drug goes in, so only first-order elimination remains, starting from the end-of-infusion level.

ALGEBRA

$$k = \frac{0.693}{3 \text{ hr}} = 0.231 \text{ hr}^{-1}\text{; } C_p = 4.45 \times e^{-0.231 \times 3} = 4.45 \times e^{-0.693} = 4.45 \times 0.500 = 2.225 \text{ mg/L}$$

Three hours is one half-life of 3 hours, so $e^{-0.693}$ is one half: half of the 4.45 mg/L from part a is left.

ROUND

$C_p = 2.2$ mg/L

2.225 mg/L to one decimal place, her value. It is half of 4.45 mg/L because 3 hours after the infusion stops is exactly one half-life of 3 hours.

> **DOES THE NUMBER MAKE SENSE?**
> Three hours is one half-life, so half of 4.45 mg/L is left: 2.2 mg/L.

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## Slides 25 and 26 Worked by the drill

**Answer:** 2.2 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **Cend** | 4.45 mg/L | part a: the end level of each infusion |
| **second start** | 6 hours after the start | infusion 2 runs 6 to 8 hr |
| **t** | 4 hours after cessation | the level at 12 hr |
| **t½** | 3 hours | $k = 0.231$ hr$^{-1}$ |
| **tinf** | 2 hour infusion | each infusion runs 2 hr |

### THE WORKING, ONE LINE AT A TIME

SETUP

Infusion 1 runs 0 to 2 hr; infusion 2 runs 6 to 8 hr; the level is asked at 8 + 4 = 12 hr

Draw the time line first. At 12 hr the first infusion has been declining for 12 − 2 = 10 hr and the second for 12 − 8 = 4 hr.

ALGEBRA

Infusion 1: $4.45 \times e^{-0.231 \times 10} = 4.45 \times 0.0993 = 0.442$ mg/L

Each identical 2-hour infusion ends at 4.45 mg/L, from part a. After it stops, $C = C_{end} e^{-kt}$, t counted from its own end.

ALGEBRA

Infusion 2: $4.45 \times e^{-0.231 \times 4} = 4.45 \times 0.3969 = 1.766$ mg/L

Four hours after the second infusion stopped, a little over one half-life of 3 hours, so a little under half of 4.45 mg/L is left.

ALGEBRA

$C_p = 0.442 + 1.766 = 2.208$ mg/L

Superposition: the concentrations from separate infusions add, because each is eliminated by first order whatever else is present.

ROUND

$C_p = 2.2$ mg/L

2.208 to one decimal place, her value. It comes out close to part b by coincidence of the times, not because the second infusion adds nothing.

> **DOES THE NUMBER MAKE SENSE?**

The second infusion alone leaves a little under half of 4.45 mg/L after 4 hours; the first adds 0.442 mg/L: 2.2 mg/L.

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## Slide 25 Worked by the drill

**Answer:** 12 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **R** | 50 mg/hr | the numerator of $C_{ss}$ |
| **t½** | 3 hours | $k = 0.231\ hr^{-1}$ |
| **$V_D$** | 18 L | with k gives Cl = 4.158 L/hr |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R}{Cl} = \frac{R}{kV_D}$$

At steady state the rate in equals the rate out, so the level is the infusion rate over the clearance. The clearance is built from k and $V_D$.

ALGEBRA

$$k = \frac{0.693}{3\ hr} = 0.231\ hr^{-1};\ C_{ss} = \frac{50\ mg/hr}{0.231 \times 18\ L} = \frac{50}{4.158} = 12.025\ mg/L$$

k is 0.693 over t½ for first-order elimination, and $Cl = kV_D$: $hr^{-1} \times L$ gives a clearance in L/hr. $C_{ss}$ is R over Cl, and mg/hr over L/hr leaves mg/L.

ROUND

$$C_{ss} \approx 12\ mg/L$$

12.025 mg/L to the nearest whole number, as she rounds it. Whatever the rate, an infusion reaches steady state in 3 to 5 half-lives of 3 hours, 9 to 15 hours.

### DOES THE NUMBER MAKE SENSE?

Cl is 0.231 × 18, 4.158 L/hr, a little over 4 L/hr, so 50 mg/hr over it is a little under 50 ÷ 4: about 12 mg/L.

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## Slide 25 Worked by the drill

**Answer:** 24 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **R** | 100 mg/hr | the new numerator |
| **t½** | 3 hours | $k = 0.231\ hr^{-1}$ |
| **$V_D$** | 18 L | with k gives Cl = 4.158 L/hr |
| **$C_{ss}$ at 50 mg/hr** | 12 mg/L | part d: doubles with the rate |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R}{Cl}, \text{ with } Cl = 0.231 \times 18 = 4.158 \text{ L/hr unchanged}$$

$C_{ss}$ is R over Cl. The clearance, k × $V_D$ from the 3-hour half-life and the 18 L volume, belongs to the drug and the patient, not to the infusion rate, so only R changes.

ALGEBRA

$$C_{ss} = \frac{100 \text{ mg/hr}}{4.158 \text{ L/hr}} = 24.05 \text{ mg/L}$$

mg/hr over L/hr leaves mg/L. With the clearance fixed, $C_{ss}$ is proportional to R, so twice the rate of part d gives twice the steady-state concentration of 12 mg/L.

ROUND

$$C_{ss} \approx 24 \text{ mg/L}$$

24.05 to the nearest whole number, her value. The time to reach it is the same 3 to 5 half-lives as at 50 mg/hr.

> **DOES THE NUMBER MAKE SENSE?**
> Doubling the rate from 50 to 100 mg/hr doubles $C_{ss}$: twice 12 mg/L is 24 mg/L.

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## Slide 25 Worked by the drill

**Answer:** C. The same time

Changing the rate changes only the concentration. The climb follows $1 - e^{-kt}$, so either rate reaches steady state in 3 to 5 half-lives of 3 hours, 9 to 15 hours.

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